

MCS-213

The Software Engineering

Q-1 Assume that you are assigned responsibility of developing an online Admit Card Generation System (OACGS) for a university. OACGS should run both on PCs and Mobile Devices. OACGS will have all fields such as student's name, student's Address, Examination Center code and Address, Examination Center Superintendent Name, Examination Center e-mail address and Mobile number, Aadhar Number, Course codes and Titles in whom the student is permitted for appearing in exams along with dates and timings of respective exams, color image of the student etc. The

student will apply for examinations online by filling examination form. The validity of the data entered by the student shall be authenticated online and OACRS consisting of the above mentioned information needs to be generated. Make assumption wherever necessary. For developing OACRS as specified above,

(a) which SDLC paradigm, will be selected, you may also suggest a SDLC paradigm that is proposed by you and non-existent as on date.

=> For developing the online Admit Card Creation system (OACRS) with the specified requirements, the most suitable Software development life cycle (SDLC) paradigm would be incremental Model.

The incremental Model is chosen for its iterative and incremental

mature, allowing the development process to be divided into smaller, manageable increments or modules. In this case, each increment can focus on a specific set of functionable or features of the ORERS.

Justification for choosing the incremental model:

1. ^{**} Early Delivery : ^{**}

The incremental model facilitates early delivery of a usable product. This is crucial in the context of an online Admit Card Generation system where users, in this case, students, would benefit from the systems as soon as certain key features are implemented.

Such as form submission and data validation.

(b) list the functional and non-functional requirements.

Ams-

let's list some of the functional and non-functional for the online admit card generations system (ONCCS) based on the provided information.

Functional Requirements:

1. ** User Registration: **

- The system should allow students to create accounts with unique usernames and passwords

2. ** online Examination Form **:

- students should be able to fill out and submit examinations forms online.
- The form should include fields for students name.

3. ** Data validation: **

- The system should authenticate and validate the entered data against predefined criteria.

4. ** Course Selection: **

- Students should be able to select the courses they want to appear for in the exams.

5. ** Examination Schedule: **

- The system should display the dates and timings of respective exams based on the selected courses.

6. ** Image upload: **

- Students should be able to upload color image of themselves for identification purpose.

7. ** Admit Card Generation: **

Non-Functional Requirements:

1. ** Performance: **

- The system should be capable of handling of a large number of simultaneous users during peak times without significant performance degradation.

2. ** Security: **

- The system must ensure the confidentiality and integrity of student data, especially sensitive information like Aadhar number.

3. ** Usability: **

- The user interface should be intuitive and user-friendly to ensure easy navigation for both PC and mobile device users.

4. ** Reliability: **

- The system should be highly reliable, with minimal downtime, or errors during critical processes like data validation and admit card generation.

5. ** Scalability: **

- The system should be scalable to accommodate an increasing number of students and examination centers.

6. ** Compatibility **

- The OACs should be compatible with various web browsers and mobile devices to ensure accessibility for a diverse user base.

7. ** Data Backup **

- Regular backups of student data.

(C) Estimate Cost.

=) Estimating the cost of developing an online Admit Card Generation System (OACGS) involves considering various factors, including development efforts, resources, Technology stack, and any additional costs.

Keep in mind that these are rough estimates and actual costs can vary based on specific projects details, geographical location, and market rates.

Development Management: **

1. ** Project Management: **

- Project planning, coordination, and management efforts.

2. ** System Analysis and Design: **

- Requirement analysis, system

design, and architecture planning.

3. ** User Interface (UI) and User Experience (UX) Design: **

- Designing an intuitive and user-friendly interface.

4. Backend Development: **

- Implementing server-side logic, database design, and integration.

5. Frontend Development: **

- Developing the user interface for both PC and mobile services.

6. ** Testing **

- Quality assurance, testing and bug fixing.

7. ** Deployment **

Resources:-

1. ** Development Team: **

- Developers, designers, and testers.

2. ** Project Manager **:

- overseeing the development process.

3. ** Database Administration **:

- Managing the database structure and performance.

4. ** System Analyst **:

- Analyzing and refining system requirements.

Technology Stack:

1. ** Hosting and Infrastructure: **

- costs associated with hosting the

application or server.

2. ** Database: **

- Licensing and maintenance costs for the database system.

3. ** Development Tools and Software **

- costs for development tools, IDEs, and software licenses.

4. ** Security Measures: **

- Implementation of security features and tools.

Additional Costs:

1. ** Training: **

- Training for end users and support staff.

2. ** Documentation: **

(d) Estimate effort.

=> Estimating the effort required to develop an online Admit Card generation System (ONACS) involving breaking down the project into various tasks and estimating the time required for each task.

Here's rough breakdown of effort estimation for different aspects of the project:

Development Efforts:

1. ** Project Management **:

- Project planning, coordination, and management:

S- 10% of the total efforts

2. ** System Analysis and Design: **

2. ** System Analysis and Design! **

- Requirement analysis, system design, and architecture planning:

10-15% of the effort.

3. ** User interface (UI) and User Experiences (UX) Design: **

- Designing an intuitive and user-friendly interface: 10-15% of the total effort.

4. ** Backend Development: **

- Implementing server-side logic, database design, and integration:

20-30% of the total effort.

5. ** Frontend Development: **

- Developing the user interface for both PC and mobile

services

20-25% of the total effort.

6. ** Integration with External System **

- Implementing authentication mechanisms and integrating with external systems for data validation.

10-15% of the total effort.

7. ** Testing : **

- Quality assurance, testing, and bug fixing.

15-20% of the total effort.

8. ** Deployment **

- Deploying the system and ensuring it is accessible to users.

5-10% of the total effort.

(e) Develop SRS using IEEE format.

=> Creating a complete Software Requirements Specification (SRS) document using IEEE format can be extensive.

Below is a simplified and condensed example of an SRS for the online Admit Card Generation system (ORACS).

Please note that a comprehensive SRS would typically include more details and sections.

Software Requirement Specification.

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed description of the requirements for the development of the online.

Admit Card Generation System (OACGS)

1.2 Scope

OACGS is intended to streamline the online examination from submission process, authentication of student data, and generation of admit cards. The system will be accessible to both PC and mobile device users.

1.3 Definitions, Acronyms, and Abbreviations

- OACGS: online Admit Card Generation System.

- UI: user interface

- UX: user experience.

- ---

2. overall Description

2.1 Product Perspective

OACGS will be a standalone system interacting with university data base for authentication and examination details. It will not be integrated with other university systems.

2.2 Product Features

- User registration and login
- online examination form submission
- Data validation and authentication
- course selection
- examination schedule display
- Admit Card generation

- Email and SMS notification
- Image upload functionality
- Mobile responsiveness

3. System Features

3.1 User Management

3.1.1 User Registration

- Users can create accounts with unique usernames and passwords

3.1.2 User Authentication

- Secure login mechanism for user authentication.

3.2 Examination Form Submission

3.2.1 Form Fields

- The system will capture student details.

(f.) list queries for whom Reports can be generated.

-) To generate reports for the Online Admit Card Generation System (OACGS), various queries can be formulated to extract valuable information from the system's database. Below are example queries that could be used to generate different types of reports:

1. ** Student information Report: **

- Query to retrieve detailed information about a specific student

'''
Sql

```
SELECT * FROM students WHERE  
Student ID = [Student ID]
```

'''

2. ** Examination Center Report: **

- Query to list all examination centers along with their details

''' Sql

```
SELECT * FROM Examination Centers;
```

3. ** Course-wise Admit Card Report **

- Query to generate admit cards for student based on selected courses.

''' Sql

```
SELECT * FROM Admit Cards WHERE  
courses Code IN (C Course Codes))
```

4. ** Examination Schedule Report: **

- Query to display the schedule for all upcoming exams.

''' Sql

```
SELECT * FROM ExamSchedule where  
Exam Date  
> = CURDATE();
```

'''

S. *** Invalid Admit Card Submission
Report: **

- Query to identify and list sub-
mission with invalid or im-
complete data.

''' Sql

```
SELECT * FROM Admit Card Submissions  
WHERE validation Status = 'Invalid'
```

6. ** Admit Cards with Image Report **

- Query to retrieve admit cards with
uploaded student images.

''' Sql.

- (g) list specific requirement which enables OACGS to run on both PCs and Mobile Devices.

Ans- To ensures that the Online Admit Card Generations System (OACGS) can run seamlessly on both PCs and Mobile Devices, specific requirement need to be addressed. Below are specific requirements tailored for cross-device compatibility:

Functional Requirements:

1. ** Responsive User interface: **

- The system's user interface should adapt and be responsive to different screen sizes, ensuring an optimal viewing experience on both PCs and various mobile devices.

2. ** Browser Compatibility: **

- The application should be compatible

with major web browsers (e.g. Chrome, Firefox, Safari) on both desktop and mobile platforms.

3. ** Mobile - Friendly Navigation: **

- Implement mobile - friendly navigation elements and controls to enhance usability and smaller screens.

4. ** Touchscreen Support: **

- Ensure that the user interface is designed to support touchscreen interactions on mobile devices.

5. ** offline Capability: **

- Provide offline capabilities or cached modes for users to access essential features even when they have limited or no internet connectivity, especially on mobile devices.

6. ** Cross-Device Data Synchronization: **

- Enables seamless data synchronization across device so that users can start a task on one device and continue on another without data loss.

7. ** Device Orientation Handling: **

- Support both portrait and landscape orientations for optimal user experience on mobile devices.

8. ** Adaptive image Handling: **

- Implement adaptive image loading to optimize image sizes based on the device's screen resolution and bandwidth.

Non-Functional Requirements:

1. ** Performance on Various Networks: **

- Optimize the system's performance

to operate efficiently on different networks speeds, especially considering mobile devices that may have varying connectivity.

2. ** Low Bandwidth consumption: **

- Minimize the use of data to reduce bandwidth consumption, particularly on mobile networks.

3. ** Cross- Platform Compatibility: **

-) Ensure that the ORACS is compatible with common operating systems such as windows, macOS, iOS and Android.

4. ** Load-Time optimization: **

-) Optimize page load times to ensure a smooth user experience, especially on mobile devices with potentially slower internet connections.